Function Observation

Tables used: STUDENT, FACULTY, DEPT, COURSE, ENROLLMENT.

AIM: To create and use user-defined functions (UDFs) in MySQL that return a single value.

Like procedures, functions need DELIMITER changed while being created (the body has semicolons in it). Change it back to ; afterwards.

-- 1. FUNCTION: Returns a single computed value

```
DELIMITER //
CREATE FUNCTION GetFacultyCount(deptName VARCHAR(10))
RETURNS INT
DETERMINISTIC
BEGIN
    DECLARE cnt INT;
    SELECT COUNT(*) INTO cnt FROM FACULTY WHERE Dept = deptName;
    RETURN cnt;
END //
DELIMITER ;

SELECT GetFacultyCount('CSE') AS CSE_Faculty_Count;
```

```
mysql> DELIMITER //
mysql> CREATE FUNCTION GetFacultyCount(deptName VARCHAR(10))
    -> RETURNS INT
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE cnt INT;
    ->     SELECT COUNT(*) INTO cnt FROM FACULTY WHERE Dept = deptName;
    ->     RETURN cnt;
    -> END //
Query OK, 0 rows affected (0.04 sec)

mysql> DELIMITER ;
mysql>
mysql> SELECT GetFacultyCount('CSE') AS CSE_Faculty_Count;
+-------------------+
| CSE_Faculty_Count |
+-------------------+
|                 3 |
+-------------------+
1 row in set (0.00 sec)

mysql>
```

-- 2. FUNCTION used inline inside SELECT (like a computed column)

```
DELIMITER //
CREATE FUNCTION AgeGroup(age INT)
RETURNS VARCHAR(10)
DETERMINISTIC
BEGIN
    IF age < 50 THEN
        RETURN 'Junior';
    ELSE
        RETURN 'Senior';
    END IF;
END //
DELIMITER ;

SELECT FacName, Age, AgeGroup(Age) AS Category FROM FACULTY;
```

```
mysql> DELIMITER //
mysql> CREATE FUNCTION AgeGroup(age INT)
    -> RETURNS VARCHAR(10)
    -> DETERMINISTIC
    -> BEGIN
    ->     IF age < 50 THEN
    ->         RETURN 'Junior';
    ->     ELSE
    ->         RETURN 'Senior';
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>
mysql> SELECT FacName, Age, AgeGroup(Age) AS Category FROM FACULTY;
+---------+------+----------+
| FacName | Age  | Category |
+---------+------+----------+
| Ramesh  |   48 | Junior   |
| Mohan   |   59 | Senior   |
| Sunitha |   46 | Junior   |
| Kavya   |   68 | Senior   |
| Arjun   |   51 | Senior   |
+---------+------+----------+
5 rows in set (0.00 sec)
```

-- 3. FUNCTION with a numeric calculation

```
DELIMITER //
CREATE FUNCTION YearsToRetire(age INT)
RETURNS INT
DETERMINISTIC
BEGIN
    RETURN 60 - age;
END //
DELIMITER ;

SELECT FacName, Age, YearsToRetire(Age) AS YearsLeft FROM FACULTY;
```

```
mysql> DELIMITER //
mysql> CREATE FUNCTION YearsToRetire(age INT)
    -> RETURNS INT
    -> DETERMINISTIC
    -> BEGIN
    ->     RETURN 60 - age;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER ;
mysql>
mysql> SELECT FacName, Age, YearsToRetire(Age) AS YearsLeft FROM FACULTY;
+---------+------+-----------+
| FacName | Age  | YearsLeft |
+---------+------+-----------+
| Ramesh  |   48 |        12 |
| Mohan   |   59 |         1 |
| Sunitha |   46 |        14 |
| Kavya   |   68 |        -8 |
| Arjun   |   51 |         9 |
+---------+------+-----------+
5 rows in set (0.00 sec)
```

-- 4. FUNCTION with a string operation

```
DELIMITER //
CREATE FUNCTION ShortName(fullName VARCHAR(50))
RETURNS VARCHAR(20)
DETERMINISTIC
BEGIN
    RETURN SUBSTRING(fullName, 1, 3);
END //
DELIMITER ;

SELECT Name, ShortName(Name) AS Abbrev FROM STUDENT;
```

```
mysql> DELIMITER //
mysql> CREATE FUNCTION ShortName(fullName VARCHAR(50))
    -> RETURNS VARCHAR(20)
    -> DETERMINISTIC
    -> BEGIN
    ->     RETURN SUBSTRING(fullName, 1, 3);
    -> END //
Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER ;
mysql>
mysql> SELECT Name, ShortName(Name) AS Abbrev FROM STUDENT;
+---------------+--------+
| Name          | Abbrev |
+---------------+--------+
| Aravind Kumar | Ara    |
| Divya         | Div    |
| Karthik       | Kar    |
| Meena         | Mee    |
| Suresh        | Sur    |
+---------------+--------+
5 rows in set (0.02 sec)
```

-- 5. FUNCTION used inside WHERE clause

```
-- Faculty who are within 5 years of retirement
SELECT FacName, Age FROM FACULTY WHERE YearsToRetire(Age) <= 5;
```

```
mysql> -- Faculty who are within 5 years of retirement
mysql> SELECT FacName, Age FROM FACULTY WHERE YearsToRetire(Age) <= 5;
+---------+------+
| FacName | Age  |
+---------+------+
| Mohan   |   59 |
| Kavya   |   68 |
+---------+------+
2 rows in set (0.00 sec)
```

-- 6. Viewing and dropping functions

```
SHOW FUNCTION STATUS WHERE Db = 'lab';
SHOW CREATE FUNCTION GetFacultyCount;
DROP FUNCTION IF EXISTS GetFacultyCount;
```

```
mysql> SHOW FUNCTION STATUS WHERE Db = 'lab';
+------+-----------------+----------+----------------+---------------------+---------------------+---------------+---------+----------------------+----------------------+--------
------------+
| Db   | Name            | Type     | Definer        | Modified            | Created             | Security_type | Comment | character_set_client | collation_connection | Databas
e Collation |
+------+-----------------+----------+----------------+---------------------+---------------------+---------------+---------+----------------------+----------------------+--------
------------+
| lab  | AgeGroup        | FUNCTION | root@localhost | 2026-08-08 14:12:36 | 2026-08-08 14:12:36 | DEFINER       |         | cp850                | cp850_general_ci     | utf8mb4
_0900_ai_ci |
| lab  | GetFacultyCount | FUNCTION | root@localhost | 2026-08-08 14:11:07 | 2026-08-08 14:11:07 | DEFINER       |         | cp850                | cp850_general_ci     | utf8mb4
_0900_ai_ci |
| lab  | ShortName       | FUNCTION | root@localhost | 2026-08-08 14:13:32 | 2026-08-08 14:13:32 | DEFINER       |         | cp850                | cp850_general_ci     | utf8mb4
_0900_ai_ci |
| lab  | YearsToRetire   | FUNCTION | root@localhost | 2026-08-08 14:13:10 | 2026-08-08 14:13:10 | DEFINER       |         | cp850                | cp850_general_ci     | utf8mb4
_0900_ai_ci |
+------+-----------------+----------+----------------+---------------------+---------------------+---------------+---------+----------------------+----------------------+--------
------------+
4 rows in set (0.02 sec)
```

```
mysql> SHOW CREATE FUNCTION GetFacultyCount;
+-----------------+------------------------------------------------------------------------------------------
-----------------------------------------------------------------------------------------------------------
---------+----------------------+----------------------+--------------------+
| Function        | sql_mode

         | character_set_client | collation_connection | Database Collation |
+-----------------+------------------------------------------------------------------------------------------
-----------------------------------------------------------------------------------------------------------
---------+----------------------+----------------------+--------------------+
| GetFacultyCount | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_E
TION `GetFacultyCount`(deptName VARCHAR(10)) RETURNS int
    DETERMINISTIC
BEGIN
    DECLARE cnt INT;
    SELECT COUNT(*) INTO cnt FROM FACULTY WHERE Dept = deptName;
    RETURN cnt;
END | cp850                | cp850_general_ci     | utf8mb4_0900_ai_ci |
+-----------------+------------------------------------------------------------------------------------------
-----------------------------------------------------------------------------------------------------------
---------+----------------------+----------------------+--------------------+
1 row in set (0.02 sec)

mysql> DROP FUNCTION IF EXISTS GetFacultyCount;
Query OK, 0 rows affected (0.01 sec)
```